

Project #2

assign October 31, 2022 due November 4, 2022

Apply Gaussian lowpass and highpass filters to two images, *kid* and *fruit* image. Assume cutoff frequency $D_0 = 100$ pixels (based on the original image size) for lowpass and highpass filters.

**Upload your project report
to E3 Assignment before 23:59 of due date.**

Your report (Word or pdf format) should contain:

- a) (20%) Source codes
- b) (20%) Fourier magnitude spectra (in Log scale) of *kid* and *fruit* image (600×600 DFT)
- c) (20%) Magnitude responses of Gaussian LPF and HPF (1200×1200 DFT)
- d) (20%) 4 output images (20%)
- e) (20%) Tables of top 25 DFT frequencies (u,v) of b) in the left half frequency region ($0 \leq u \leq M-1$, $0 \leq v \leq N/2-1$)

Use 150dpi to plot the images in problems b) – d).

Kid.tiff
600x600



Fruit.tiff
600x600

